

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Subject: 341151 Electric Circuits I

Date: 7 October 2014

Year: 2014

Section: 5 - 6

Time: 15.00 - 17.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks and dictionaries are **NOT** allowed.
4. Calculator is allowed.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. The examination has 10 pages (including this page), 3 parts (54 questions) and a total score of 70 points.
7. Write all your answers on this examination sheet.

Part 1 (20 points) True/False

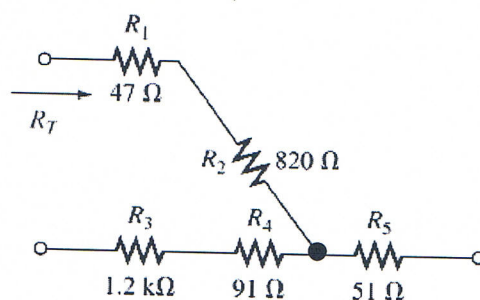
Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. From English standard, MKS represents Meters, Kilometers, and Seconds.
2. The SI unit of inductance is named after Michael Faraday.
3. The notation Ms is an abbreviation for micro seconds.
4. Coulomb's law is about force between charges.
5. Coulomb is the unit of electric charge in SI unit.
6. For simple atom structure, nucleus contains protons and electrons.
7. Secondary cell is the battery that cannot be recharged.
8. The unit of conductance is in Ohm.
9. Gold is a better conductor than copper.
10. The shorter electric wire length, the lower resistance.
11. For four-band resistor's color code, the third band represents power-of-ten multiplier.
12. Power in watt is defined by Joule per second.
13. Efficiency can be determined by input power over output power.
14. Kilowatt-hour meter is used to measure the energy of electricity supplied to the residential user.
15. The current that flows through series elements of a circuit is the same in each element.
16. KVL stands for Kelvin Voltage Law.
17. Voltage divider is a method for finding electric current in two parallel resistors.
18. One horse power is equivalent to 764 watts.
19. Electric power can be obtained from I^2R .
20. Potential between "short terminals" is equal to zero volt.

Part 2 (30 points) Multiple choices

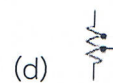
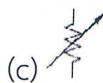
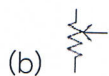
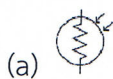
Instruction: Mark the correct answer for following questions in the provided answer sheet (LAST PAGE).

- Evaluate the following expression: $(1 \times 10^{-9}) / (1 \times 10^{-6}) =$
 (a) 0.01 (b) 1×10^{-3}
 (c) 100 (d) 1×10^3
- Express the number 0.000001 using the powers of ten.
 (a) 1×10^{-4} (b) 1×10^{-5}
 (c) 1×10^{-6} (d) 1×10^{-7}
- Three 12-volt batteries connected together by a conductor, positive terminal to negative terminal will result in which of the following voltage?
 (a) 12 volts (b) 24 volts
 (c) 36 volts (d) 48 volts
- The current divider rule states that for two _____ resistors of equal value, the current will divide equally.
 (a) series (b) parallel
 (c) disconnected (d) none
- An ammeter is always placed _____ an element to measure the current.
 (a) in series with (b) in front of
 (c) in parallel with (d) on top of
- From Figure below, the total resistance R_T is



- $47\Omega + 820\Omega + 51\Omega$
 - $51\Omega + 91\Omega + 1.2k\Omega$
 - $47\Omega + 820\Omega + 91\Omega + 1.2k\Omega$
 - $47\Omega + 820\Omega + 51\Omega + 91\Omega + 1.2k\Omega$
- What is the power deliver to a resistor $10k\Omega$ with voltage across of 10 volts?
 (a) 1 mW (b) 10 mW (c) 100 mW (d) 1 W

8. What is the symbol of potentiometer?



9. Which of the following is the list of conductors from higher conductivity to lower conductivity?

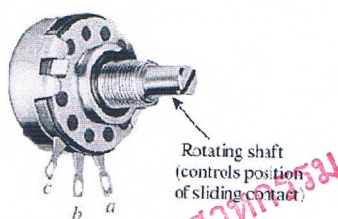
(a) Silver – Copper – Gold – Aluminum

(b) Gold – Silver – Copper – Aluminum

(c) Silver – Gold – Aluminum – Copper

(d) Gold – Silver – Aluminum – Copper

10. From figure below, which terminal pair will not affect the change in resistance if use ohm meter to measure.



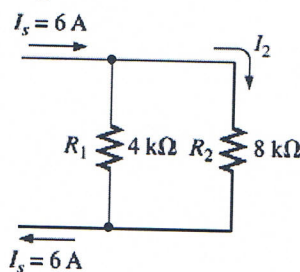
(a) a and b

(b) a and c

(c) b and c

(d) none

11. From figure below, the value of I_2 is



(a) 0 A

(b) 2 A

(c) 4 A

(d) 6 A

12. How much charge passes through a battery of 1 V if the energy expended is 10 J?

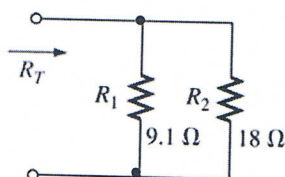
(a) 0.1 C

(b) 1 C

(c) 10 C

(d) 100 C

13. From figure below, what R_T should be?



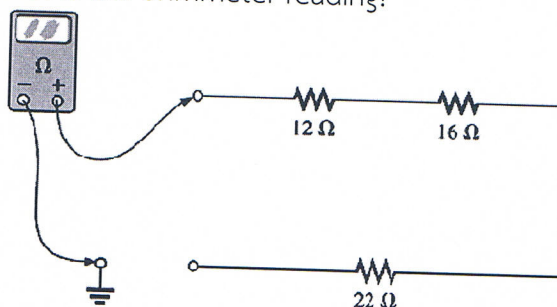
(a) Lower than 9.1 ohms.

(b) Slightly lower than 18 ohms.

(c) Higher than 18 ohms.

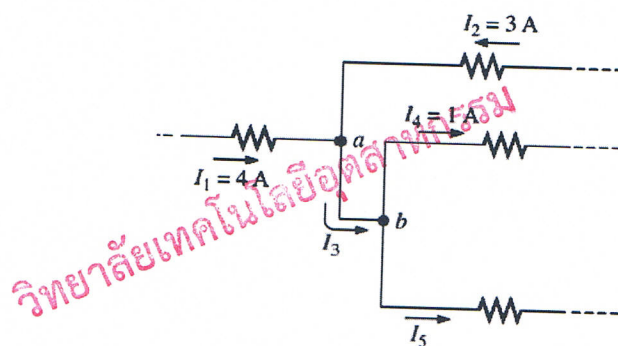
(d) Lower than 18 ohms but slightly higher than 9.1 Ω.

14. From figure below, what is the ohmmeter reading?



- (a) 0 Ohms
(b) 50 Ohms
(c) 12 Ohms
(d) Infinity Ohms

15. The value of I_3 is _____ and I_5 is _____.



- (a) - 7 A and - 6 A
(b) 7 A and - 6 A
(c) - 7 A and 6 A
(d) 7 A and 6 A

16. Potential is defined as the _____ at a point with respect to another point in the electrical system.

- (a) charge
(b) current
(c) ampere
(d) voltage

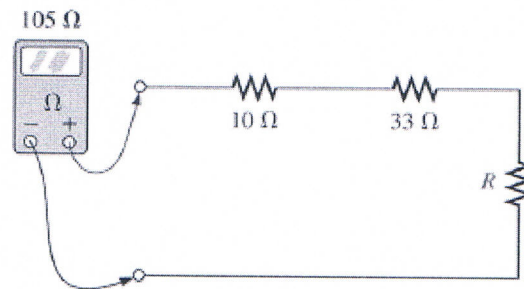
17. If a resistor uses 420 J of energy for 7 min, what is the power to the resistor?

- (a) 1 mW
(b) 80 mW
(c) 1 W
(d) 80 W

18. Which of the following expresses a double-superscript notation of $V_{ab} = -30$ V?

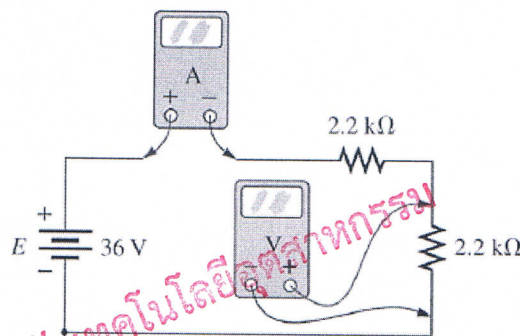
- (a) Point a has a higher potential than b.
(b) Point a has the same potential as b.
(c) Point a has a lower potential than b.
(d) Point a cannot be determined.

19. From Figure below, the value of R is



- (a) 43Ω (b) 62Ω (c) 82Ω (d) 148Ω

20. From Figure below, which of the following gives the value of voltmeter reading?



- (a) 12V (b) 18V (c) 36V (d) 48V

21. From previous problem, which of the following gives the value of ammeter reading?

- (a) 0 mA (b) 2.2 mA (c) 8.18 mA (d) 16.36 mA

22. A circuit has a source voltage of 15 volts and two series resistors with a total resistance of 4000 ohms. If R_1 has a potential of 7.5 volts, what is the value of R_2 ?

- (a) 1000 ohms (b) 2000 ohms
(c) 4000 ohms (d) 8000 ohms

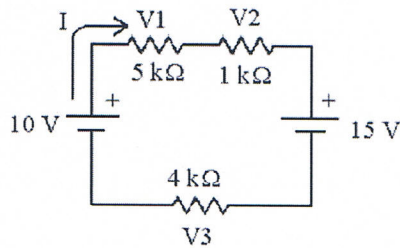
23. For how many hours will a battery with an Ah rating of 32 theoretically provide a current of 1.75 A?

- (a) 17.3 h (b) 18.3 h (c) 19.3 h (d) 20.3 h

24. A 250 W light bulb operates on 120 V. How much current is flowing through the bulb?

- (a) 20.8 S (b) 2.08 mA (c) 0.208 S (d) 2.08 A

25. See figure below, which one of the following KVL equations describes this circuit?

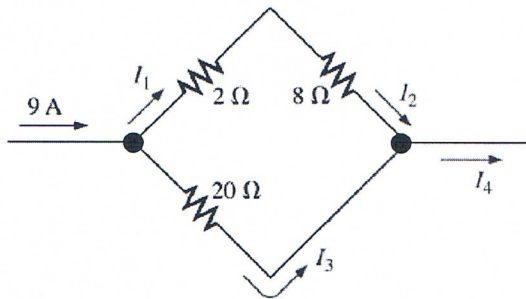


- (a) $+10\text{ V} + V1 + V2 - 15\text{ V} + V3 = 0$
 (b) $+10\text{ V} - V1 - V2 + 15\text{ V} + V3 = 0$
 (c) $-10\text{ V} - V1 - V2 - 15\text{ V} - V3 = 0$
 (d) $-10\text{ V} + V1 + V2 + 15\text{ V} + V3 = 0$
26. What is the cost of using a 24 W radio for 1 hour at a rate of 2 baht per kilowatthour?
 (a) 0.024 baht (b) 0.24 baht (c) 0.048 baht (d) 0.48 baht
27. For parallel elements, the total conductance is the _____ of the individual conductance.
 (a) algebraic sum (b) parallel total (c) series total (d) sum
28. A wattmeter is an instrument used to measure _____.
 (a) power (b) current (c) resistance (d) voltage
29. The current divider rule is similar to the voltage divider rule except that the current _____ the resistances.
 (a) divides in an inverse relationship to the values of
 (b) divides in a direct relationship to the values of
 (c) always divides equally among
 (d) divides independent of the values of
30. What is the efficiency of a motor that has an output of 1.5 hp with an input of 1550 W?
 (a) 0.722 (b) 7.22 (c) 0.139 (d) 1.39

Part 3 (20 points) Calculation

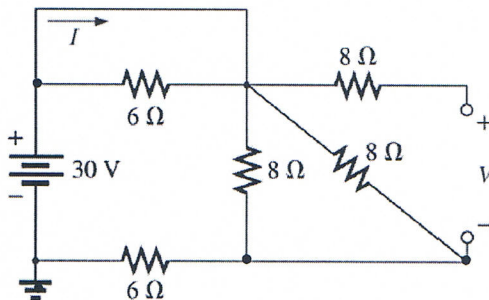
Instruction: Shows the mathematical expression and answers of following problems.

1. From figure below, determine the value current I_1 , I_2 , I_3 , and I_4

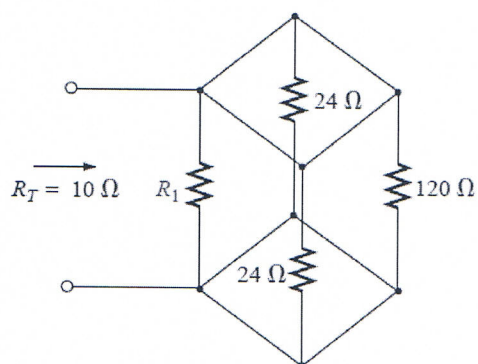


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2. From Figure below, find the voltage V and current I .

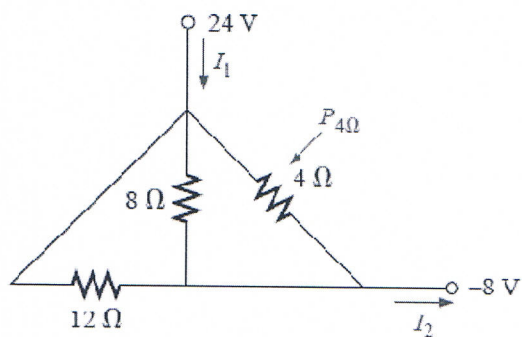


3. Find the value of R_1 .



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4. Determine power $P_{4\Omega}$ and electric current I_1 and I_2 .



Answer Sheet

Name: _____ ID: _____

Subject: 341151 Electric Circuits I

Part1

	True	False
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Part 2

	a	b	c	d
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